US 20250283734A1

(19) United States
(12) Patent Application Publication
(10) Pub. No.: US 2025/0283734 A1
(43) Pub. Date: Sep. 11, 2025
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(54) **POSITION DETECTION DEVICE AND ACTUATOR**

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(21) Appl. No.: **19/068,269**

(22) Filed: **Mar. 3, 2025**

(30) **Foreign Application Priority Data**

Mar. 8, 2024 (JP) 2024-036084

Publication Classification

(51) **Int. Cl.**
G01D 5/14 (2006.01)
H02J 50/00 (2016.01)

(52) **U.S. Cl.**
CPC ***G01D 5/14*** (2013.01); ***H02J 50/00*** (2016.02)

(57) **ABSTRACT**

A position detection device includes a first sensor that outputs a first detection signal when a piston is located within a first detection range, a detection unit that detects the piston position based on the first detection signal, an antenna that receives power wirelessly, a power supply line through which the power is supplied from the antenna to the first sensor when the piston is located within the first detection range, and a power output line configured to output, when the piston is located outside the first detection range, the power to a sensor module used for detecting the piston position.

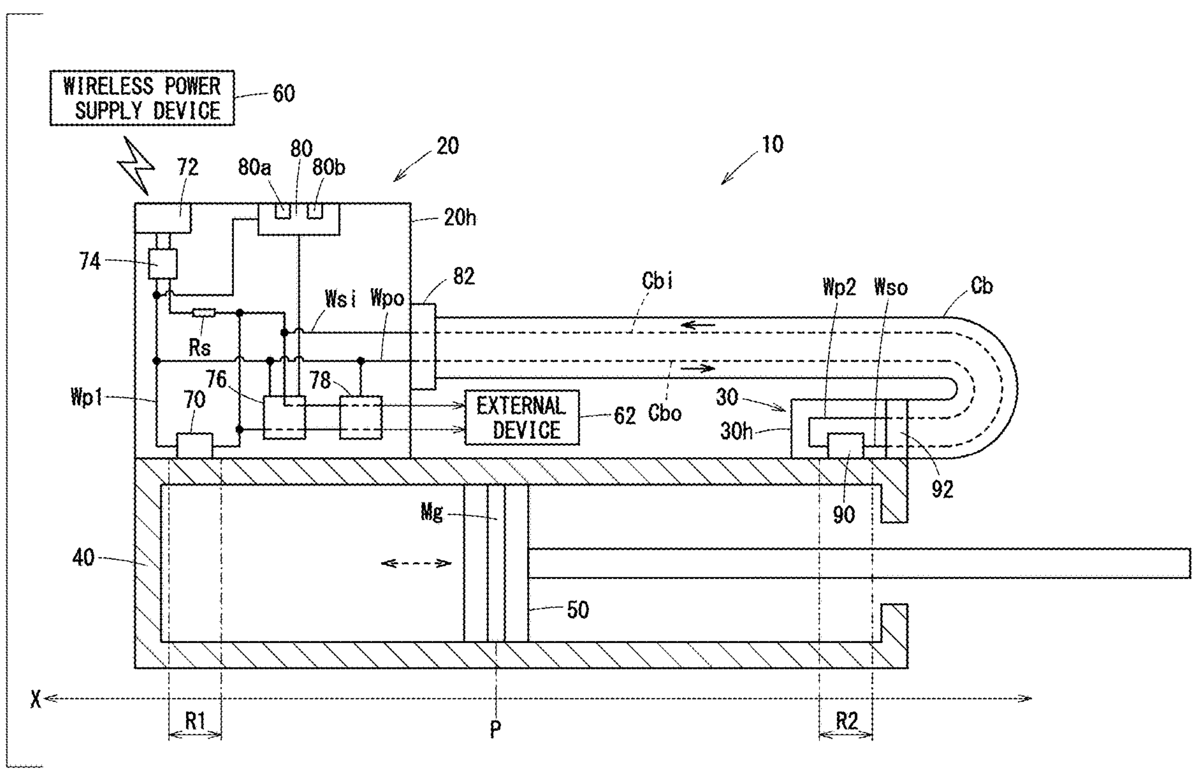


FIG. 1

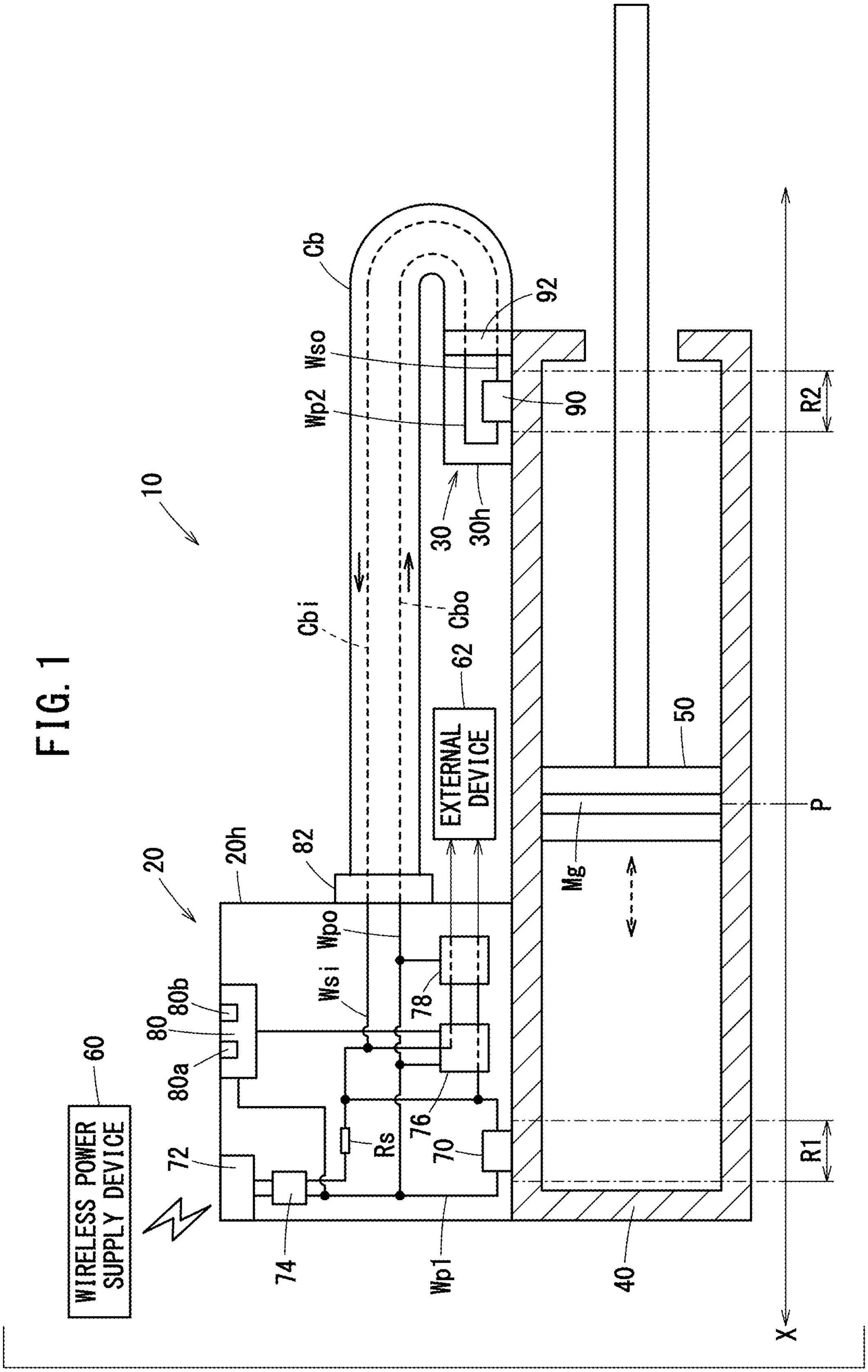
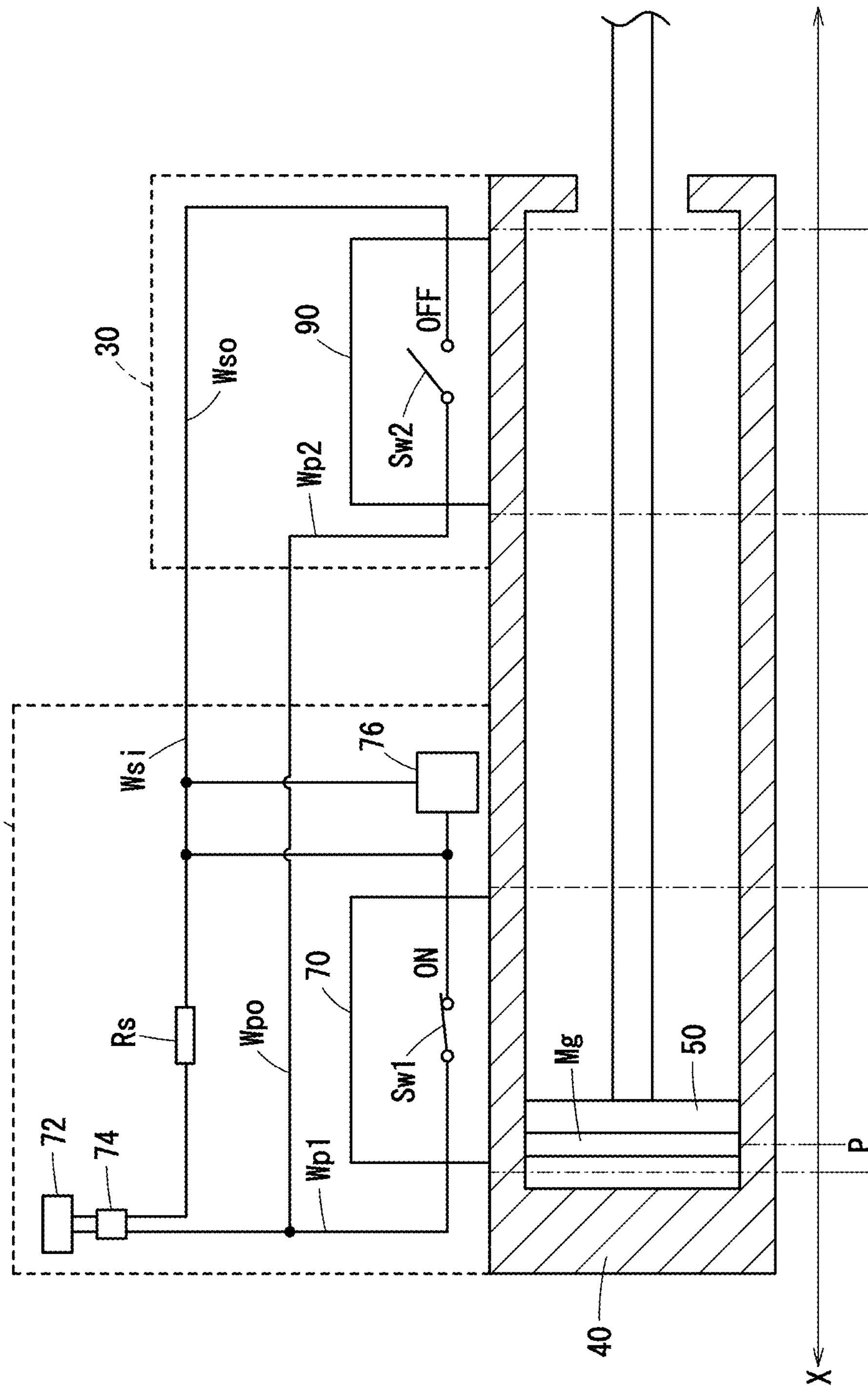

WIRELESS POWER SUPPLY DEVICE
EXTERNAL DEVICE
10
20
20h
30
30h
40
50
60
62
70
72
74
76
78
80
80a
80b
82
90
92
Cb
Cbi
Cbo
Wso
Wp2
Wpo
Wsi
Wp1
Rs
Mg
P
R1
R2
X

FIG. 2



20
30
72
74
Rs
Wsi
Wso
Wpo
Wp1
Wp2
70
76
90
ON
OFF
Sw1
Sw2
Mg
50
40
P
R1
R2
X

FIG. 3

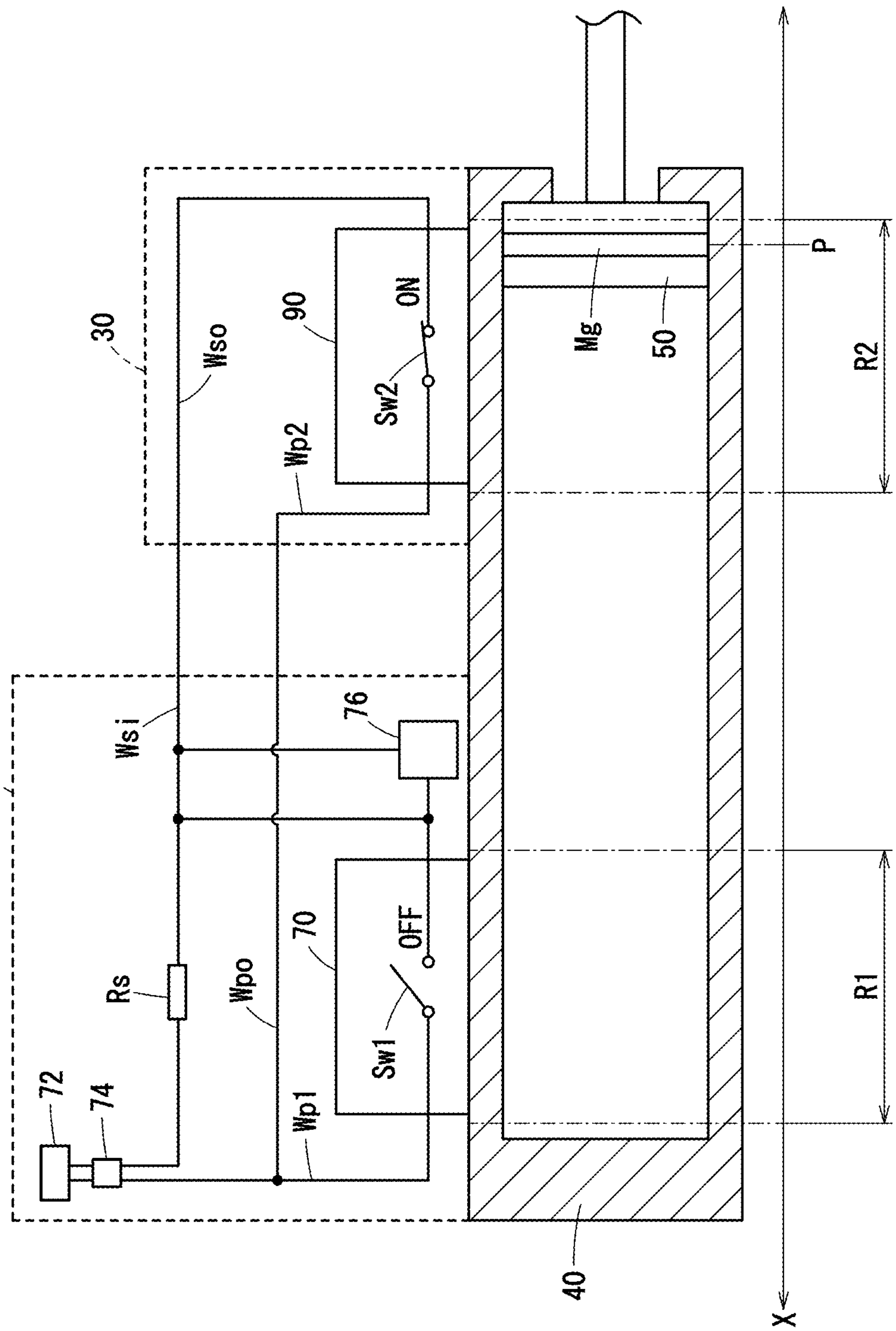

20
30
72
74
Rs
Wpo
Wp1
70
Sw1
OFF
76
Ws i
Wso
90
ON
Sw2
Wp2
Mg
50
P
R2
R1
40
X

FIG. 4

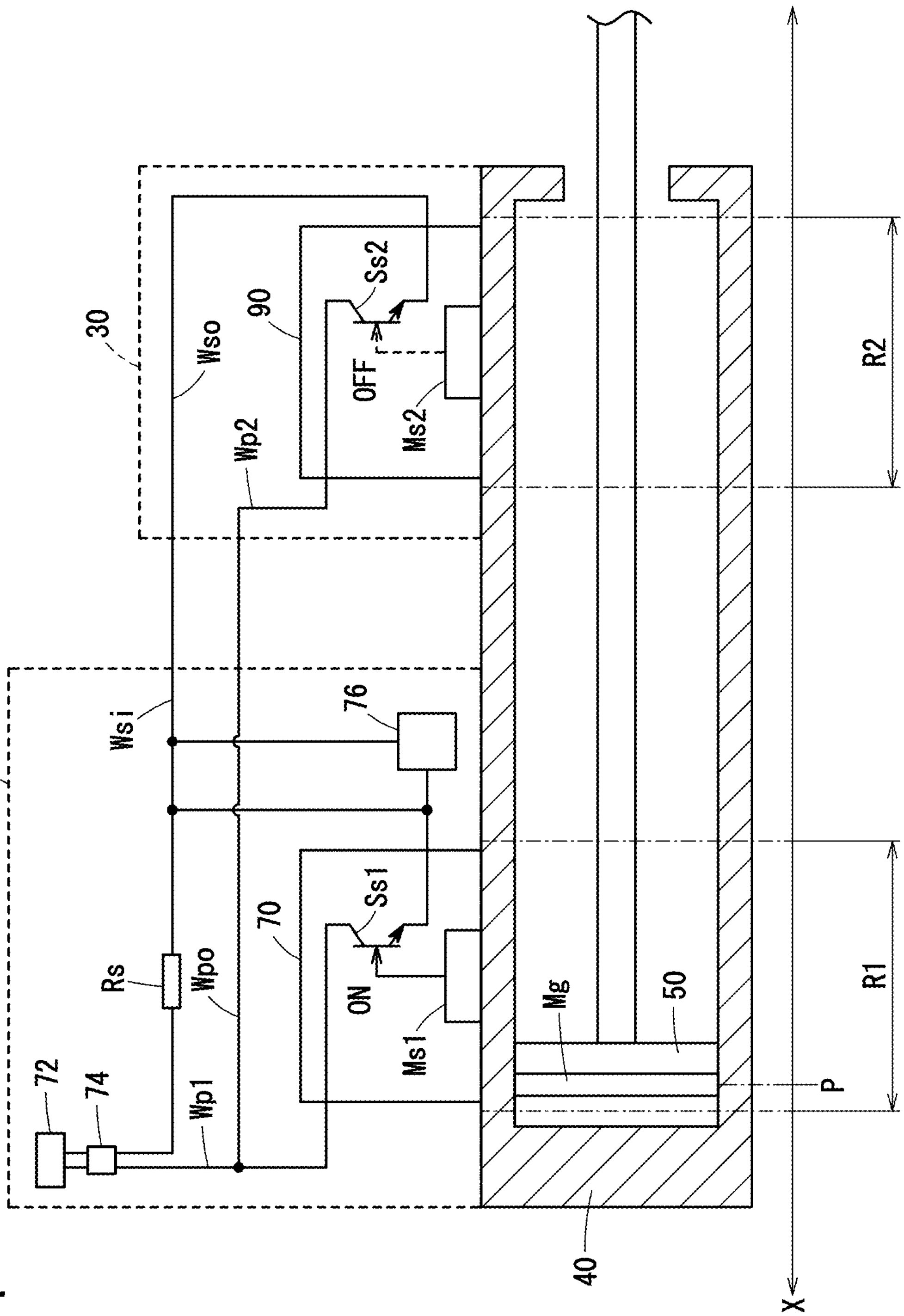
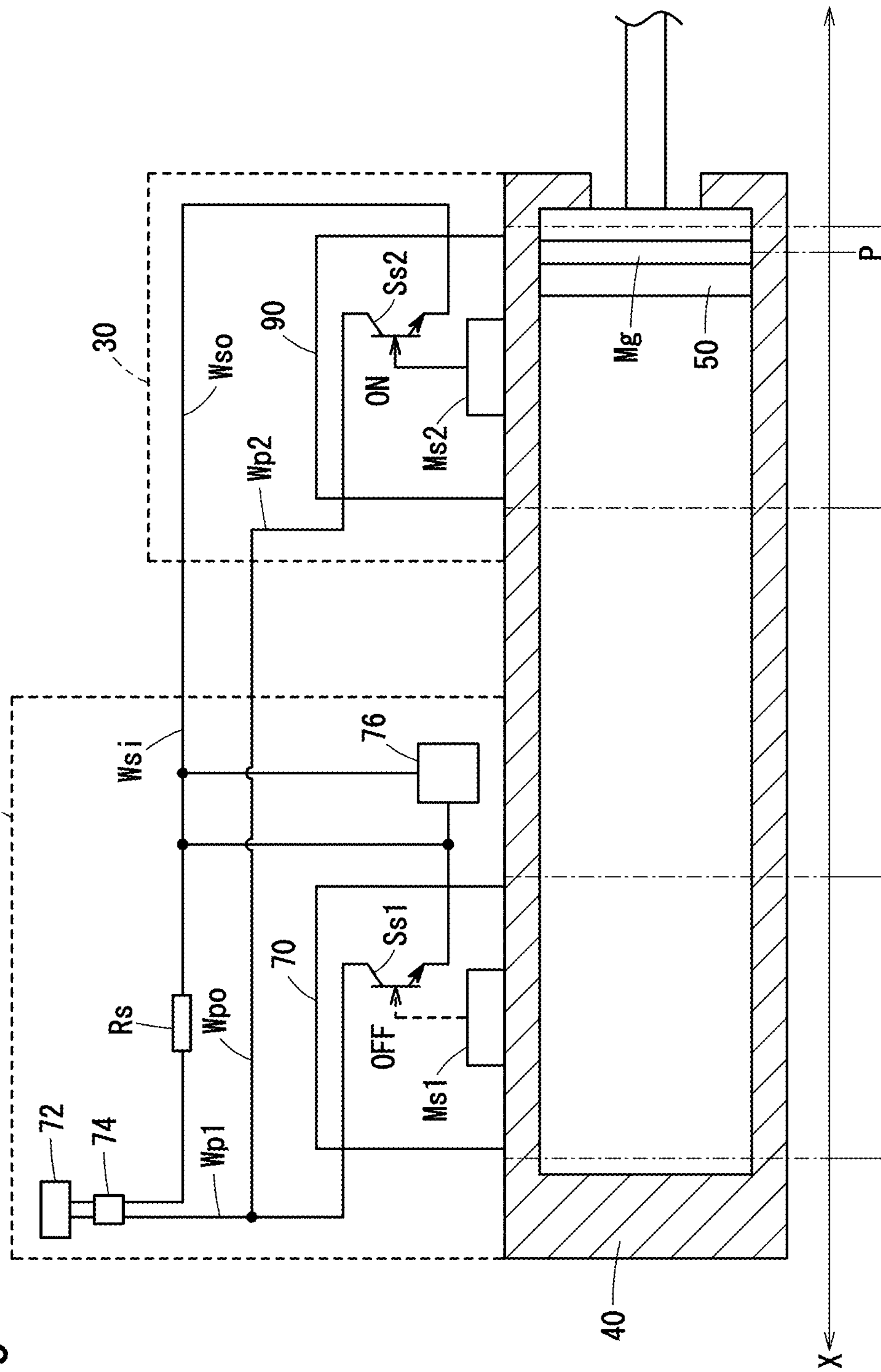

30
Wso
90
Ss2
OFF
Ms2
Wp2
R2
20
Wsi
76
Rs
Wpo
70
Ss1
ON
Ms1
Mg
50
P
R1
72
74
Wp1
40
X

FIG. 5


30
Wso
90
Ss2
ON
Ms2
Wp2
Mg
50
P
R2
20
Wsi
76
Ss1
70
Rs
Wpo
OFF
Ms1
72
74
Wp1
R1
40
X

POSITION DETECTION DEVICE AND ACTUATOR

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-036084 filed on Mar. 8, 2024, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present invention relates to a position detection device and an actuator.

Description of the Related Art

[0003] JP 2023-078294 A discloses a wireless antenna module for communication. Electric power is supplied to the first and second module bodies, which are two wireless antenna modules, from a battery that stores wirelessly supplied power. The first module body wirelessly transmits a sensor detection signal from a first device body to a master device (GW unit). The second module body wirelessly transmits a sensor detection signal from a second device body to the master device (GW unit).

SUMMARY OF THE INVENTION

[0004] The wirelessly supplied power is often weak. Therefore, if weak power is supplied to both of the two wireless antenna modules, a power shortage may occur in any of the wireless antenna modules.

[0005] The present invention has the object of solving the aforementioned problems.

[0006] A first aspect of the present invention is a position detection device for detecting a piston position of a piston that is attached to a cylinder and moves in the cylinder, comprising: a first sensor that outputs a first detection signal in a case where the piston is located within a first detection range; a detection unit that detects the piston position based on the first detection signal; a communication device that transmits the piston position detected by the detection unit to an external device; an antenna that receives power wirelessly; a power supply line through which the power is supplied from the antenna to the first sensor in a case where the piston is located within the first detection range; and a power output line configured to externally output, in a case where the piston is located outside the first detection range, the power to a sensor module used for detecting the piston position.

[0007] A second aspect of the present invention is an actuator that includes the position detection device according to the first aspect, the sensor module, the cylinder, and the piston.

[0008] The present invention can reduce the possibility of the shortage of wirelessly supplied power.

[0009] The above and other objects, features, and advantages of the present invention will become more apparent from the following description when taken in conjunction with the accompanying drawings, in which a preferred embodiment of the present invention is shown by way of illustrative example.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. **1** is a diagram illustrating an actuator;

[0011] FIG. **2** is a diagram for explaining a configuration example of a first sensor and a second sensor, and power supply to the first sensor and the second sensor according to a piston position;

[0012] FIG. **3** is a diagram for explaining the configuration example of the first sensor and the second sensor, and power supply to the first sensor and the second sensor according to a piston position;

[0013] FIG. **4** is a diagram for explaining another configuration example of a first sensor and a second sensor, and power supply to the first sensor and the second sensor according to a piston position; and

[0014] FIG. **5** is a diagram for explaining another configuration example of the first sensor and the second sensor, and power supply to the first sensor and the second sensor according to a piston position.

DETAILED DESCRIPTION OF THE INVENTION

[0015] FIG. **1** is a diagram illustrating an actuator **10**. The actuator **10** is used for conveying a workpiece or the like and has a position detection device **20**, a sensor module **30**, a cylinder **40**, and a piston **50**. The position detection device **20** and the sensor module **30** are both mounted on the outer surface of the cylinder **40** and are connected to each other through a cable Cb.

[0016] The piston **50** moves in the X direction in the cylinder **40**. That is, the moving direction of the piston **50** is the X direction. The position detection device **20** detects a piston position P of the piston **50** in the X direction. The piston **50** is provided with a magnet Mg. In the example shown in FIG. **1**, the piston position P corresponds to the position of the magnet Mg.

[0017] The position detection device **20** receives power from the wireless power supply device **60** wirelessly. The position detection device **20** and the sensor module **30** are used for detecting the piston position P. When the piston **50** moves in the cylinder **40** in the X direction, a first detection signal or a second detection signal regarding the piston position P of the piston **50** may be output from the position detection device **20** or the sensor module **30**. The output of the first detection signal or the second detection signal is performed using the power received by the position detection device **20** wirelessly. The position detection device **20** detects the piston position P based on the first detection signal or the second detection signal.

[0018] The position detection device **20** transmits the detected piston position P to an external device **62**. The external device **62** is, for example, a management device that manages the position detection device **20** or the actuator **10**. The transmission of the piston position P from the position detection device **20** to the external device **62** is performed by wireless communication or wired communication.

[0019] The configurations of the position detection device **20** and the sensor module **30** will be further described with reference to FIG. **1**. The position detection device **20** includes a first sensor **70**, an antenna **72**, a power conversion circuit **74**, a detection unit **76**, a communication device **78**, an indicator **80**, and a connector **82**. The first sensor **70**, the antenna **72**, the power conversion circuit **74**, the detection unit **76**, the communication device **78**, and the indicator **80** are housed in a housing **20***h* of the position detection device **20**. The connector **82** is provided at the housing **20***h* in such a way that part of the connector **82** is exposed on the outer surface of the housing **20***h*.

[0020] The sensor module **30** has a second sensor **90** and a connector **92**. The second sensor **90** is housed in a housing **30***h* of the sensor module **30**. The connector **92** is provided at the housing **30***h* in such a way that part of the connector **92** is exposed on the outer surface of the housing **30***h*.

[0021] Compared to the position detection device **20**, the sensor module **30** includes significantly fewer components. Therefore, the sensor module **30** is extremely smaller than the position detection device **20**. Therefore, an area occupied by the sensor module **30** when the sensor module **30** is attached to the cylinder **40** can be much smaller than the position detection device **20**.

[0022] The antenna **72** of the position detection device **20** receives power from the wireless power supply device **60** wirelessly. The power conversion circuit **74** converts the power received by the antenna **72**. The power conversion circuit **74** converts, for example, AC power into DC power. The power converted by the power conversion circuit **74** is supplied to the first sensor **70**, the detection unit **76**, the communication device **78**, and the indicator **80**. Line connections for the power supplied to the detection unit **76**, the communication device **78**, and the indicator **80** are schematically illustrated in FIG. **1**.

[0023] The position detection device **20** further has a power supply line Wp**1**. The power supply line Wp**1** connects the power conversion circuit **74** and the first sensor **70**. When the magnet Mg of the piston **50** is located within a first detection range R**1** due to the movement of the piston **50** in the X direction, the piston position P is located within the first detection range R**1**. In that case, the first sensor **70** is turned on. The configuration of the first sensor **70** will be described later with reference to FIGS. **2** to **5**.

[0024] When the first sensor **70** is turned on, power is supplied from the antenna **72** to the first sensor **70** via the power conversion circuit **74** through the power supply line Wp**1**. The power supplied to the first sensor **70** is consumed at a load resistor Rs. Thus, the voltage output from the first sensor **70** is obtained as the above-described first detection signal output from the first sensor **70**. That is, when the magnet Mg of the piston **50** is located within the first detection range R**1**, the first sensor **70** outputs the first detection signal.

[0025] The detection unit **76** includes an electronic circuit such as a CPU (Central Processing Unit). The detection unit **76** acquires the first detection signal output from the first sensor **70**. The detection unit **76** detects the piston position P based on the first detection signal. That is, the detection unit **76** detects that the piston position P is located within the first detection range R**1**. The communication device **78** transmits the piston position P detected by the detection unit **76** to the external device **62**. The communication between the communication device **78** and the external device **62** may be wireless communication or wired communication.

[0026] The indicator **80** has a first lamp **80***a* and a second lamp **80***b* that emit light in different colors. The power wirelessly supplied from the wireless power supply device **60** may be weak. To suppress the power consumption, the first lamp **80***a* and the second lamp **80***b* emit light intermittently. When the piston position P is detected based on the first detection signal by the detection unit **76**, the indicator **80** turns on the first lamp **80***a* and then turns it off. The indicator **80** makes the first lamp **80***a* blink by repeatedly turning on and off the first lamp **80***a*. That is, the first lamp **80***a* intermittently emits light.

[0027] The position detection device **20** further has a power output line Wpo. One end of the power output line Wpo is connected to the power conversion circuit **74** whereas the other end of the power output line Wpo is connected to the connector **82**. The cable Cb connecting the position detection device **20** and the sensor module **30** can be attached to the connector **82**. When the magnet Mg of the piston **50** is located outside the first detection range R**1** and the second sensor **90** in the sensor module **30** is ON, the power output line Wpo can externally output to the sensor module **30** the power received by the antenna **72** and converted by the power conversion circuit **74**. The externally output power may be supplied to the second sensor **90** as described later.

[0028] As described above, when the magnet Mg of the piston **50** is located within the first detection range R**1**, the power received by the antenna **72** is supplied to the first sensor **70** through the power supply line Wp**1**. When the magnet Mg of the piston **50** is located outside the first detection range R**1** and the second sensor **90** is on, the power received by the antenna **72** can be externally output to the sensor module **30** through the power output line Wpo and supplied to the second sensor **90**.

[0029] That is, wasteful power supply is prevented in which the wirelessly supplied power is supplied to both the first sensor **70** and the second sensor **90**. This can reduce the possibility of the shortage of wirelessly supplied power.

[0030] The above-described cable Cb can be attached to the connector **92** of the sensor module **30**. In the example shown in FIG. **1**, one end of the cable Cb is attached to the connector **82** of the position detection device **20** whereas the other end of the cable Cb is attached to the connector **92** of the sensor module **30**. The cable Cb houses an output line Cbo extending from the position detection device **20** to the sensor module **30** and an input line Cbi extending from the sensor module **30** to the position detection device **20**. The above-mentioned power output line Wpo is connected, via the connector **82**, to the output line Cbo housed in the cable Cb.

[0031] The sensor module **30** further has a power supply line Wp**2**. The power supply wiring Wp**2** connects the connector **92** with the second sensor **90**. The power supply wiring Wp**2** is connected, via the connector **92**, to the output line Cbo housed in the cable Cb. When the magnet Mg of the piston **50** is located within the second detection range R**2** due to the movement of the piston **50** in the X direction, the piston position P is located within the second detection range R**2**. In that case, the second sensor **90** is turned on. The configuration of the second sensor **90** will be described later with reference to FIGS. **2** to **5**.

[0032] When the second sensor **90** is turned on, power is supplied from the antenna **72** of the position detection device **20** to the second sensor **90** via the power conversion circuit **74** through the power output line Wpo and the connector **82** of the position detection device **20**, the output line Cbo in the cable Cb, and the connector **92** and the power supply line Wp**2** of the sensor module **30**. The voltage output from the second sensor **90** is obtained as the above-described second detection signal output from the second sensor **90**. That is, when the magnet Mg of the piston **50** is located within the second detection range R**2**, the second sensor **90** outputs the second detection signal.

[0033] The sensor module **30** further has a signal output wiring Wso. The signal output wiring Wso connects the second sensor **90** and the connector **92**. The signal output wiring Wso can externally output the second detection signal to the position detection device **20** via the connector **92** and the cable Cb. The signal output wiring Wso is connected, via the connector **92**, to the input line Cbi housed in the cable Cb.

[0034] The position detection device **20** further has a signal input wiring Wsi. The signal input wiring Wsi connects the connector **82** and the detection unit **76**. The signal input wiring Wsi is connected, via the connector **82**, to the input line Cbi housed in the cable Cb. The second detection signal output from the second sensor **90** of the sensor module **30** is supplied to the detection unit **76** through the signal input wiring Wsi.

[0035] The detection unit **76** acquires the second detection signal output from the second sensor **90**. The detection unit **76** detects the piston position P based on the second detection signal. That is, the detection unit **76** detects that the piston position P is located within the second detection range R**2**. The communication device **78** transmits the piston position P detected by the detection unit **76** to the external device **62**. When the piston position P is detected based on the second detection signal by the detection unit **76**, the indicator **80** turns on the second lamp **80***b* and then turns it off. The indicator **80** makes the second lamp **80***b* blink by repeatedly turning on and off the second lamp **80***b*. That is, the second lamp **80***b* intermittently emits light.

[0036] The signal input wiring Wsi is also connected to the load resistor Rs. Thus, the power supplied to the second sensor **90** is consumed at the load resistance Rs.

[0037] The first sensor **70** is placed at a position corresponding to one end of the stroke of the piston **50** moving in the X direction, for example. The second sensor **90** is, for example, arranged at a position corresponding to the other end of the stroke of the piston **50** described above. That is, the first sensor **70** of the position detection device **20** is attached to the cylinder **40** at a position apart from the second sensor **90** of the sensor module **30** along the X direction. The first sensor **70** and the second sensor **90** are positioned in a manner so that the first detection range R**1** of the position detection device **20** and the second detection range R**2** of the sensor module **30** do not overlap in the X direction.

[0038] Therefore, when the magnet Mg of the piston **50** is located within the first detection range R**1**, the first sensor **70** is on while the second sensor **90** is off because the magnet Mg of the piston **50** is not located within the second detection range R**2**. In this case, the first sensor **70** outputs the first detection signal but the second sensor **90** does not output the second detection signal. Unnecessary power supply to the second sensor **90** is prevented.

[0039] When the magnet Mg of the piston **50** is located within the second detection range R**2**, the second sensor **90** is on while the first sensor **70** is off because the magnet Mg of the piston **50** is not located within the first detection range R**1**. In this case, the second sensor **90** outputs the second detection signal but the first sensor **70** does not output the first detection signal. Unnecessary power supply to the first sensor **70** is prevented. Thus, there is not a case where both the first sensor **70** and the second sensor **90** are turned on. This can further reduce the possibility of the shortage of wirelessly supplied power.

[0040] When the magnet Mg of the piston **50** is located outside the first detection range R**1** and outside the second detection range R**2**, both the first sensor **70** and the second sensor **90** are turned off.

[0041] When the first detection signal is output from the first sensor **70**, the detection unit **76** detects the piston position P based on the first detection signal. When the piston position P is detected based on the first detection signal by the detection unit **76**, the indicator **80** turns on the first lamp **80***a*. In this case, the second detection signal is not output, and thus the second lamp **80***b* is not turned on.

[0042] When the second detection signal is output from the second sensor **90**, the detection unit **76** detects the piston position P based on the second detection signal. When the piston position P is detected based on the second detection signal by the detection unit **76**, the indicator **80** turns on the second lamp **80***b*. In this case, the first detection signal is not output, and thus the first lamp **80***a* is not turned on. Thus, the piston position P can be detected easily and the piston position P can be notified to a worker who performs an operation using the actuator **10**.

[0043] FIGS. **2** and **3** are diagrams for explaining the configuration example of the first sensor **70** and the second sensor **90** and also explaining the power supply to the first sensor **70** and the second sensor **90** according to the piston position P. The illustration of the line connection for the power supplied to the detection unit **76**, of the communication device **78**, of the indicator **80**, and the like is omitted. Both the first sensor **70** and the second sensor **90** shown in FIGS. **2** and **3** are reed switches. Specifically, the first sensor **70** is a reed switch Sw**1**. The second sensor **90** is a reed switch Sw**2**. The reed switches Sw**1** and Sw**2** open and close in response to the magnetism of the magnet Mg provided on the piston **50**.

[0044] In the example shown in FIG. **2**, the piston position P of the piston **50** is included in the first detection range R**1**. The first detection range R**1** of the first sensor **70** is determined based on the piston position P given when the reed switch Sw**1** is closed by the magnetism of the magnet Mg of the piston **50**. When the piston **50** is located within the first detection range R**1**, the reed switch Sw**1** is closed and turned on.

[0045] Thus, power is supplied from the antenna **72** to the reed switch Sw**1** via the power conversion circuit **74** and the power supply line Wp**1**. The power supplied to the reed switch Sw**1** is consumed at the load resistor Rs. The voltage is output from the reed switch Sw**1** that is turned on. That is, the reed switch Sw**1** outputs the first detection signal. The detection unit **76** detects the piston position P based on the first detection signal output from the reed switch Sw**1**.

[0046] The first detection range R**1** of the first sensor **70** of the position detection device **20** and the second detection range R**2** of the second sensor **90** of the sensor module **30** do not overlap in the X direction. Therefore, in the example shown in FIG. **2**, the piston position P is not included in the second detection range R**2**. The reed switch Sw**2**, which can be closed by the magnetism of the magnet Mg of the piston **50**, is not closed. That is, when the piston **50** is located outside the second detection range R**2**, the reed switch Sw**2** is open and off. Therefore, no power is supplied from the antenna **72** to the reed switch Sw**2**. The reed switch Sw**2** does not output the second detection signal.

[0047] In the example shown in FIG. **3**, the piston position P of the piston **50** is included in the second detection range R**2**. The second detection range R**2** of the second sensor **90** is determined based on the piston position P given when the reed switch Sw**2** is closed by the magnetism of the magnet Mg of the piston **50**. When the piston **50** is located within the second detection range R**2**, the reed switch Sw**2** is closed and turned on.

[0048] Thereby, power is supplied from the antenna **72** to the reed switch Sw**2** via the power conversion circuit **74**, the power output line Wpo, and the power supply line Wp**2**. The power supplied to the reed switch Sw**2** is consumed at the load resistor Rs. The voltage is output from the reed switch Sw**2** that has been turned on. That is, the reed switch Sw**2** outputs the second detection signal. The detection unit **76** detects the piston position P based on the second detection signal output from the reed switch Sw**2**.

[0049] In the example shown in FIG. **3**, the piston position P is not included in the first detection range R**1**. The reed switch Sw**1**, which can be closed by the magnetism of the magnet Mg of the piston **50**, is not closed. That is, when the piston **50** is located outside the first detection range R**1**, the reed switch Sw**1** is open and off. Therefore, no power is supplied from the antenna **72** to the reed switch Sw**1**. The reed switch Sw**1** does not output the first detection signal.

[0050] As described above, the reed switch Sw**1** is turned on when the piston **50** is located within the first detection range R**1**, and outputs the first detection signal. The reed switch Sw**1** is turned off when the piston **50** is located outside the first detection range R**1**, and does not output the first detection signal. Thus, it is possible to detect in a simple manner whether the piston position P is located within the first detection range R**1**.

[0051] As described above, the reed switch Sw**2** is turned on when the piston **50** is located within the second detection range R**2**, and outputs the second detection signal. The reed switch Sw**2** is turned off when the piston **50** is located outside the second detection range R**2**, and does not output the second detection signal. Thus, it is possible to detect in a simple manner whether the piston position P is located within the second detection range R**2**.

[0052] Switches other than the reed switches Sw**1** and Sw**2** may be used as the first sensor **70** and the second sensor **90**. FIGS. **4** and **5** are diagrams for explaining another example of the configuration of the first sensor **70** and the second sensor **90** and explaining the power supply to the first sensor **70** and the second sensor **90** according to the piston position P. The illustration of the line connection for the power supplied to the detection unit **76**, of the communication device **78**, of the indicator **80**, and the like is omitted.

[0053] Both the first sensor **70** and the second sensor **90** shown in FIGS. **4** and **5** include a magnetic sensor and a semiconductor switch. Specifically, the first sensor **70** includes a magnetic sensor Ms**1** and a semiconductor switch Ss**1**. The second sensor **90** includes a magnetic sensor Ms**2** and a semiconductor switch Ss**2**. The magnetic sensors Ms**1** and Ms**2** output electric signals corresponding to current or voltage in accordance with the magnetism of the magnets Mg provided at the piston **50**. The magnetic sensors Ms**1** and Ms**2** are, for example, magneto-resistive (MR) elements, Hall elements, or the like.

[0054] The electric signal output from the magnetic sensor Ms**1** is input to the semiconductor switch Ss**1**, whereby the semiconductor switch Ss**1** is turned on. The electric signal output from the magnetic sensor Ms**2** is input to the semiconductor switch Ss**2**, whereby the semiconductor switch Ss**2** is turned on.

[0055] In the example shown in FIG. **4**, the piston position P of the piston **50** is included in the first detection range R**1**. A first detection range R**1** of the first sensor **70** is determined based on the piston position P when the electric signal is output from the magnetic sensor Ms**1** due to the magnetism possessed by the magnet Mg of the piston **50**. When the piston **50** is located within the first detection range R**1**, the electric signal is output from the magnetic sensor Ms**1**, and the semiconductor switch Ss**1** is turned on.

[0056] Thus, power is supplied from the antenna **72** to the semiconductor switch Ss**1** via the power conversion circuit **74** and the power supply line Wp**1**. The power supplied to the semiconductor switch Ss**1** is consumed at the load resistor Rs. Voltage is output from the semiconductor switch Ss**1** that has been turned on. That is, the semiconductor switch Ss**1** outputs the first detection signal. The detection unit **76** detects the piston position P based on the first detection signal output from the semiconductor switch Ss**1**.

[0057] The first detection range R**1** of the first sensor **70** of the position detection device **20** and the second detection range R**2** of the second sensor **90** of the sensor module **30** do not overlap in the X direction. Therefore, in the example shown in FIG. **4**, the piston position P is not included in the second detection range R**2**. The electric signal that can be output from the magnetic sensor Ms**2** to the semiconductor switch Ss**2** due to the magnetism of the magnet Mg of the piston **50** is not output.

[0058] That is, when the piston **50** is located outside the second detection range R**2**, no electric signal is input from the magnetic sensor Ms**2** to the semiconductor switch Ss**2**, and the semiconductor switch Ss**2** is off. Therefore, no power is supplied from the antenna **72** to the semiconductor switch Ss**2**. The semiconductor switch Ss**2** does not output the second detection signal.

[0059] In the example shown in FIG. **5**, the piston position P of the piston **50** is included in the second detection range R**2**. The second detection range R**2** of the second sensor **90** is determined based on the piston position P given when the electric signal is output from the magnetic sensor Ms**2** due to the magnetism possessed by the magnet Mg of the piston **50**. When the piston **50** is located within the second detection range R**2**, the electric signal is output from the magnetic sensor Ms**2**, and the semiconductor switch Ss**2** is turned on.

[0060] Thus, power is supplied from the antenna **72** to the semiconductor switch Ss**2** via the power conversion circuit **74**, the power output line Wpo, and the power supply line Wp**2**. The power supplied to the semiconductor switch Ss**2** is consumed at the load resistor Rs. The voltage is output from the semiconductor switch Ss**2** that has been turned on. That is, the semiconductor switch Ss**2** outputs the second detection signal. The detection unit **76** detects the piston position P based on the second detection signal output from the semiconductor switch Ss**2**.

[0061] In the example shown in FIG. **5**, the piston position P is not included in the first detection range R**1**. The electric signal that can be output from the magnetic sensor Ms**1** to the semiconductor switch Ss**1** due to the magnetism of the magnet Mg of the piston **50** is not output. That is, when the piston **50** is located outside the first detection range R**1**, no electric signal is input from the magnetic sensor Ms**1** to the semiconductor switch Ss**1**, and the semiconductor switch Ss**1** is off. Therefore, no power is supplied from the antenna **72** to the semiconductor switch Ss**1**. The semiconductor switch Ss**1** does not output the first detection signal.

[0062] As described above, when the piston **50** is located within the first detection range R**1**, the magnetic sensor Ms**1** outputs the electric signal, and the semiconductor switch Ss**1** is turned on to output the first detection signal. When the piston **50** is located outside the first detection range R**1**, the magnetic sensor Ms**1** does not output an electric signal, and the semiconductor switch Ss**1** is turned off to not output the first detection signal. Thus, it is possible to detect in a simple manner whether the piston position P is located within the first detection range R**1**.

[0063] As described above, when the piston **50** is located within the second detection range R**2**, the magnetic sensor Ms**2** outputs the electric signal, and the semiconductor switch Ss**2** is turned on to output the second detection signal. When the piston **50** is located outside the second detection range R**2**, the magnetic sensor Ms**2** does not output an electric signal, and the semiconductor switch Ss**2** is turned off to not output the second detection signal. Thus, it is possible to detect in a simple manner whether the piston position P is located within the second detection range R**2**.

[0064] With respect to the above embodiments, we further disclose the following supplementary notes.

Supplementary Note 1

[0065] A position detection device (**20**) for detecting a piston position (P) of a piston (**50**) that is attached to a cylinder (**40**) and moves in the cylinder includes a first sensor (**70**) that outputs a first detection signal in a case where the piston is located within a first detection range (R**1**), a detection unit (**76**) that detects the piston position based on the first detection signal, a communication device (**78**) that transmits the piston position detected by the detection unit to an external device (**62**), an antenna (**72**) that receives power wirelessly, a power supply line (Wp**1**) through which the power is supplied from the antenna to the first sensor in a case where the piston is located within the first detection range, and a power output line (Wpo) configured to externally output, in a case where the piston is located outside the first detection range, the power to a sensor module (**30**) used for detecting the piston position. Such a configuration can reduce the possibility of the shortage of wirelessly supplied power.

Supplementary Note 2

[0066] The position detection device according to Supplementary note 1 may be configured in such a way that the piston is provided with a magnet (Mg), and the first sensor is a reed switch (Sw**1**) that is turned on to output the first detection signal in a case where the piston is located within the first detection range and that is turned off to not output the first detection signal in a case where the piston is located outside the first detection range. According to such a configuration, it is possible to detect in a simple manner whether the piston position is located within the first detection range.

Supplementary Note 3

[0067] The position detection device according to Supplementary note 1 may be configured in such a way that the piston is provided with a magnet, the first sensor includes a magnetic sensor (Ms**1**) that outputs an electric signal in a case where the piston is located within the first detection range, and a semiconductor switch (Ss**1**) that is turned on by input of the electric signal and outputs the first detection signal, and in a case where the piston is located outside the first detection range, the magnetic sensor does not output the electric signal, and the semiconductor switch is turned off due to absence of the input of the electric signal and does not output the first detection signal. According to such a configuration, it is possible to detect in a simple manner whether the piston position is located within the first detection range.

Supplementary Note 4

[0068] The position detection device according to Supplementary note 1 may be configured in such a way that the sensor module includes a second sensor (**90**) that outputs a second detection signal in a case where the piston is located within a second detection range (R**2**), the first sensor is attached to the cylinder at a position apart from the second sensor along the movement direction (X) of the piston, and the first sensor and the second sensor are positioned in a manner so that the first detection range and the second detection range do not overlap in the movement direction. Such a configuration can further reduce the possibility of the shortage of wirelessly supplied power.

Supplementary Note 5

[0069] The position detection device described in Supplementary note 1 may further include a connector (**82**) to which a cable (Cb) connecting the position detection device and the sensor module is attachable, wherein the power output wiring may be connected to the connector. According to such a configuration, the wirelessly supplied power can be externally output to the sensor module.

Supplementary Note 6

[0070] The position detection device according to Supplementary note 5 may further include a signal input line (Wsi) that connects the connector and the detection unit, wherein the sensor module may include a second sensor that outputs a second detection signal in a case where the piston is located within a second detection range, the second detection signal output from the second sensor may be supplied to the detection unit through the signal input line, and the detection unit may detect the piston position based on the second detection signal. According to such a configuration, it is possible to detect whether the piston position is located within the second detection range.

Supplementary Note 7

[0071] The position detection device according to Supplementary note 4 may be configured in such a way that the detection unit detects the piston position based on the first detection signal in a case where the first detection signal is output from the first sensor, and detects the piston position based on the second detection signal in a case where the second detection signal is output from the second sensor. According to such a configuration, the piston position can be easily detected.

Supplementary Note 8

[0072] The position detection device described in Supplementary note 7 may further include an indicator (**80**) that includes a first lamp (**80***a*) and a second lamp (**80***b*) that emit light in different colors, wherein the indicator may turn on the first lamp in a case where the piston position is detected based on the first detection signal by the detection unit, and the indicator may turn on the second lamp in a case where the piston position is detected based on the second detection signal by the detection unit. Thus, the piston position can be informed to a worker who performs an operation using the actuator.

Supplementary Note 9

[0073] An actuator (**10**) is provided with the position detection device described in supplementary note 4, the sensor module described in supplementary note 4, the cylinder described in supplementary note 4, and the piston described in supplementary note 4. Such a configuration can reduce the possibility of the shortage of wirelessly supplied power during an operation using the actuator.

[0074] Although the present disclosure has been detailed, the present disclosure is not limited to the individual embodiments described above. These embodiments may be variously added, replaced, altered, partially deleted, etc., without departing from the scope of the present disclosure or the intent of the present disclosure as derived from the claims and their equivalents. These embodiments can also be implemented in combination. For example, in the above-described embodiments, the order of the operations and the order of the processes are shown as an example, and are not limited to these examples. The same applies to the case where numerical values or mathematical expressions are used in the description of the above-described embodiments.

1. A position detection device for detecting a piston position of a piston that is attached to a cylinder and moves in the cylinder, the position detection device comprising:
- a first sensor that outputs a first detection signal in a case where the piston is located within a first detection range;
- a detection unit that detects the piston position based on the first detection signal;
- a communication device that transmits the piston position detected by the detection unit to an external device;
- an antenna that receives power wirelessly;
- a power supply line through which the power is supplied from the antenna to the first sensor in a case where the piston is located within the first detection range; and
- a power output line configured to externally output, in a case where the piston is located outside the first detection range, the power to a sensor module used for detecting the piston position.

2. The position detection device according to claim **1**, wherein
- the piston is provided with a magnet, and
- the first sensor is a reed switch that is turned on to output the first detection signal in a case where the piston is located within the first detection range and that is turned off to not output the first detection signal in a case where the piston is located outside the first detection range.

3. The position detection device according to claim **1**, wherein
- the piston is provided with a magnet,
- the first sensor includes a magnetic sensor that outputs an electric signal in a case where the piston is located within the first detection range, and a semiconductor switch that is turned on by input of the electric signal and outputs the first detection signal, and
- in a case where the piston is located outside the first detection range, the magnetic sensor does not output the electric signal, and the semiconductor switch is turned off due to absence of the input of the electric signal and does not output the first detection signal.

4. The position detection device according to claim **1**, wherein
- the sensor module includes a second sensor that outputs a second detection signal in a case where the piston is located within a second detection range,
- the first sensor is attached to the cylinder at a position apart from the second sensor along a moving direction of the piston, and
- the first sensor and the second sensor are positioned in a manner so that the first detection range and the second detection range do not overlap in the moving direction.

5. The position detection device according to claim **1**, further comprising
- a connector to which a cable connecting the position detection device and the sensor module is attachable,
- wherein the power output line is connected to the connector.

6. The position detection device according to claim **5**, further comprising
- a signal input line that connects the connector and the detection unit,
- wherein the sensor module includes a second sensor that outputs a second detection signal in a case where the piston is located within a second detection range,
- the second detection signal output from the second sensor is supplied to the detection unit through the signal input line, and
- the detection unit detects the piston position based on the second detection signal.

7. The position detection device according to claim **4**, wherein
- the detection unit detects the piston position based on the first detection signal in a case where the first detection signal is output from the first sensor, and detects the piston position based on the second detection signal in a case where the second detection signal is output from the second sensor.

8. The position detection device according to claim **7**, further comprising
- an indicator that includes a first lamp and a second lamp that emit light in different colors,
- wherein the indicator turns on the first lamp in a case where the piston position is detected based on the first detection signal by the detection unit, and
- the indicator turns on the second lamp in a case where the piston position is detected based on the second detection signal by the detection unit.

9. An actuator comprising:
- the position detection device according to claim **4**;
- the sensor module according to claim **4**;
- the cylinder according to claim **4**; and
- the piston according to claim **4**.

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